Bounding Boxes, Matching, and Detection Metrics

Mathematics of Deep Learning — Chapter 16

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Foundation Books Project

Roadmap

From Classification to Detection

What Object Detection Asks For

Classification: one label for a whole image. Detection: a *set* of localized objects, each with class and score.

- Ground truth: $\mathcal{G} = \{(b_j^\star, c_j^\star)\}_{j=1}^m$.
- Prediction: $f_\theta(x) = \{(\hat{b}_i, \hat{c}_i, \hat{s}_i)\}_{i=1}^{K(x)}$.
- $K(x)$ varies image to image — output is a *set*, not a vector.

Detection answers three questions at once: what, where, and how confident.

Running Example: One Street Scene

Image x of width $W = 640$, height $H = 480$. Three ground-truth objects:

Object	Box (x_1, y_1, x_2, y_2)	Role in the chapter
Person	$(80, 90, 210, 430)$	Large, easy to localize
Bicycle	$(250, 250, 500, 420)$	Wide, loose-box candidates
Traffic sign	$(500, 70, 555, 135)$	Small, sensitive to pixel error

Predictions will include a correct person, a duplicate person box, a loose bicycle, a background false positive, and a missed sign.

These are routine failure modes, not edge cases.

Boxes and Image Geometry

Axis-Aligned Bounding Boxes

An image has a coordinate system before it has a neural network. x increases to the right; y increases *downward*.

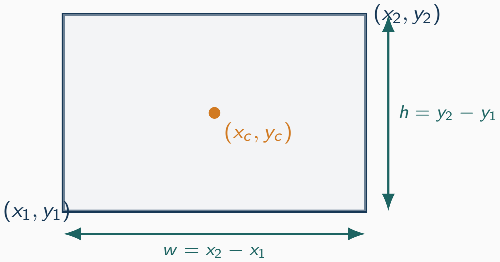
Corner form
$b = (x_1, y_1, x_2, y_2)$ with $0 \leq x_1 < x_2 \leq W$, $0 \leq y_1 < y_2 \leq H$.
Width $w(b) = x_2 - x_1$, height $h(b) = y_2 - y_1$, area $w(b)h(b)$.

Center-size form
$b = (x_c, y_c, w, h)$ with $x_c = (x_1 + x_2)/2$, $w = x_2 - x_1$.

Person box: width 130, height 340, area 44,200. Sign: area 3,575. A ten-pixel error is a bigger fraction of the sign.

State the coordinate convention before training, eval, or annotation review.

Two Forms, One Box



Conversion

$x_c = (x_1 + x_2)/2$

$y_c = (y_1 + y_2)/2$

$w = x_2 - x_1, \;\; h = y_2 - y_1$

Inverse: $x_1 = x_c - w/2$, etc.

- Corner form: convenient for intersection arithmetic.
- Center-size form: convenient for predicting a residual from a reference.
- Normalized form $(\tilde{x}, \tilde{y}) \in [0, 1]^2$ is resolution-agnostic.

Many “model” bugs are coordinate or resize bugs in disguise.

Intersection over Union

IoU: A Geometric Score for Box Overlap

For boxes A, B with positive union area,

$$\mathrm{IoU}(A, B) = \frac{\mathrm{area}(A \cap B)}{\mathrm{area}(A \cup B)} \in [0, 1].$$

Corner-form intersection:

$$\begin{aligned} w_\cap &= \max\{0, \min(x_2^A, x_2^B) - \max(x_1^A, x_1^B)\}, \\ h_\cap &= \max\{0, \min(y_2^A, y_2^B) - \max(y_1^A, y_1^B)\}. \end{aligned}$$

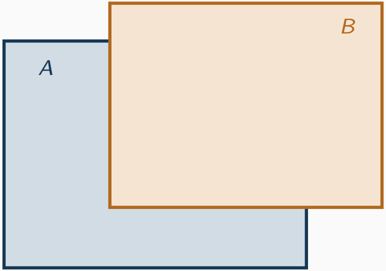
- The $\max\{0, \cdot\}$ guards against disjoint boxes.
- $\mathrm{IoU} = 0$ when disjoint; $\mathrm{IoU} = 1$ when the boxes coincide.
- IoU is geometric — not a probability and not a score.

IoU rewards shared area and penalizes extra area at the same time.

Animated IoU: Building Intersection and Union

(1) Box A alone.

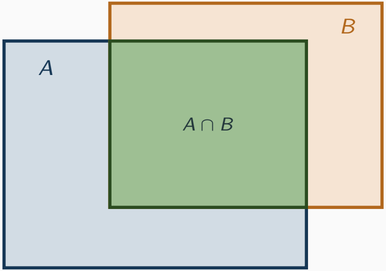
Animated IoU: Building Intersection and Union



(1) Box A alone.

(2) Box B overlaps A.

Animated IoU: Building Intersection and Union

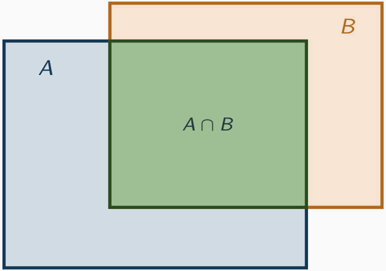


(1) Box A alone.

(2) Box B overlaps A.

(3) Green region is the intersection.

Animated IoU: Building Intersection and Union



Example: person detection
$G = (80, 90, 210, 430)$, $P = (90, 100, 220, 420)$
$w_\cap = 120$, $h_\cap = 320$
$|A \cap B| = 38{,}400$
$|A \cup B| = 44{,}200 + 41{,}600 - 38{,}400$
IoU ≈ 0.810

(1) Box A alone.
(2) Box B overlaps A.
(3) Green region is the intersection.
(4) Plug numbers into the formula.

Same recipe whether boxes barely touch or nearly coincide.

IoU Thresholds and Benchmarks

A prediction counts as a localization match when $\mathrm{IoU}(\hat{b}_i, b_j^\star) \geq \tau$.

- PASCAL VOC: AP at $\tau = 0.5$ — localization is loose.
- COCO: average mAP over $\tau \in \{0.50, 0.55, \ldots, 0.95\}$ — stricter on geometry.
- Variants: GIoU, DIoU, CIoU add geometric penalties so equal-IoU predictions can be ranked.

GIoU subtracts the slack inside the enclosing box C:

$$\mathrm{GIoU}(\hat{b}, b^\star) = \mathrm{IoU}(\hat{b}, b^\star) - \frac{|C| - |\hat{b} \cup b^\star|}{|C|}.$$

Two mAP numbers are not comparable unless the IoU convention matches.

Detection = Classification + Localization

Two Questions per Candidate

Each candidate must answer:

- **What class?** Predict $\hat{p}_i \in \Delta^C$ over C classes.
- **Where exactly?** Predict box $\hat{b}_i \in \mathbb{R}^4$.

Multitask loss for matched candidates:

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \lambda\,\mathcal{L}_{\mathrm{box}}, \qquad \lambda > 0.$$

- Too small λ: right class, sloppy boxes.
- Too large λ: tight boxes, mixed-up classes.

λ is a training-objective decision, not a cosmetic constant.

Loss Pieces: Cross-Entropy, Smooth-L_1, Focal

Classification
Cross-entropy or binary cross-entropy on $\hat{p}_i$ vs. one-hot or objectness target.

Localization: smooth L_1 on residual r

$$\mathrm{smooth}_{L_1}(r) = \begin{cases} \frac{1}{2}r^2, & |r| < 1, \\ |r| - \frac{1}{2}, & |r| \geq 1. \end{cases}$$

Focal loss (binary), $p_t = p$ **if** $y = 1$ **else** $1 - p$

$$\mathcal{L}_{\mathrm{focal}} = -\alpha(1 - p_t)^{\gamma} \log p_t.$$

Easy examples ($p_t \to 1$) contribute much smaller gradients.

Match the training loss to the eval metric, or surprises follow.

Candidate Boxes and Anchors

Anchors: Reference Shapes at Many Locations

An anchor $a = (x_a, y_a, w_a, h_a)$ is a reference box placed at a known feature-map location. The detector predicts a *correction*:

$$t_x = \frac{x - x_a}{w_a}, \quad t_y = \frac{y - y_a}{h_a}, \quad t_w = \log \frac{w}{w_a}, \quad t_h = \log \frac{h}{h_a}.$$

Inverse: $x = x_a + t_x w_a$, $w = w_a \exp(t_w)$, etc.

- Translations normalized by anchor size.
- Scale changes on a log scale (stable, multiplicative).
- Model predicts how to move and resize, not boxes from nothing.

Anchors turn absolute box regression into a normalized offset problem.

Animated: Anchor Grid and Assignment

image / feature map

(1) Start with the image plane.

Animated: Anchor Grid and Assignment

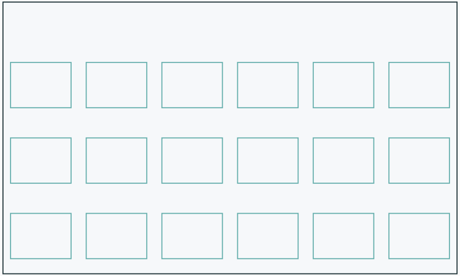


image / feature map

(1) Start with the image plane.

(2) Place a grid of anchors at many locations.

Animated: Anchor Grid and Assignment

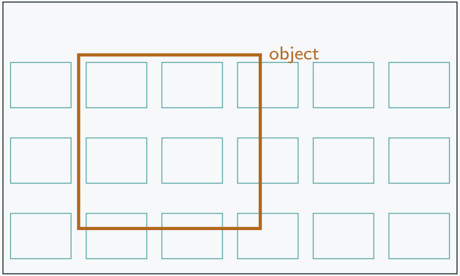


image / feature map

(1) Start with the image plane.
(2) Place a grid of anchors at many locations.
(3) Introduce a ground-truth object.

Animated: Anchor Grid and Assignment

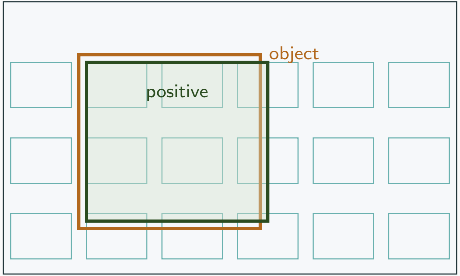


image / feature map

(1) Start with the image plane.
(2) Place a grid of anchors at many locations.
(3) Introduce a ground-truth object.
(4) Anchor with IoU $\geq \tau_+$ is positive.

Animated: Anchor Grid and Assignment

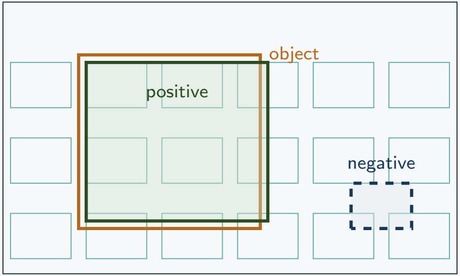


image / feature map

(1) Start with the image plane.
(2) Place a grid of anchors at many locations.
(3) Introduce a ground-truth object.
(4) Anchor with IoU $\geq \tau_+$ is positive.
(5) Anchor with IoU $< \tau_-$ is negative.

Animated: Anchor Grid and Assignment

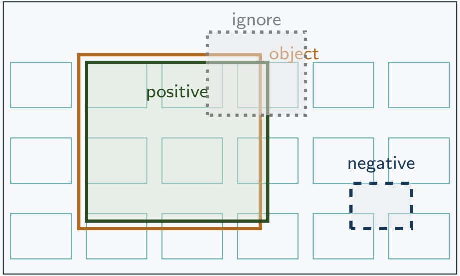


image / feature map

(1) Start with the image plane.

(2) Place a grid of anchors at many locations.

(3) Introduce a ground-truth object.

(4) Anchor with IoU $\geq \tau_+$ is positive.

(5) Anchor with IoU $< \tau_-$ is negative.

(6) Anchors in the gray zone are ignored.

Dense Prediction and Anchor-Free Alternatives

- Dense detectors attach predictions to many feature-map locations, scales, and aspect ratios.
- Feature pyramid networks predict at several resolutions for scale.
- Anchor-free detectors (FCOS, etc.) predict object centers or distances to box sides — no hand-designed anchor shapes.
- DETR uses a fixed set of learned object queries instead of a grid.

Assignment thresholds (anchor-based):

$$\text{IoU} \geq \tau_+ \Rightarrow \text{positive}, \quad \text{IoU} < \tau_- \Rightarrow \text{negative}, \quad \text{else ignored.}$$

Different families differ in candidates — they still all need matching.

Matching Predictions to Ground Truth

Matching: Connecting Two Unordered Sets

A loss or metric cannot compare two unordered sets directly.

Class-aware IoU match
Pair (i, j) qualifies if $\hat{c}_i = c_j^\star$ and $\mathrm{IoU}(\hat{b}_i, b_j^\star) \geq \tau$.

- **Training assignment**: many positives per object (dense detectors).
- **Evaluation matching**: one-to-one — each ground-truth object explains at most one true positive.
- Second prediction on the same object $\rightarrow$ duplicate false positive.

"Matched to the object" means different things at training and eval time.

A Tiny Matching Table

Three person-class predictions; one person ground truth; $\tau = 0.5$.

Prediction	Score	IoU with person	Eval result
$\hat{b}_1$	0.92	0.81	True positive
$\hat{b}_2$	0.80	0.70	Duplicate false positive
$\hat{b}_3$	0.65	0.20	False positive (low IoU)

Remark: for one image with m ground-truth objects, one-to-one evaluation matching gives at most m true positives.

One-to-one matching caps TPs — duplicates always hurt precision.

Hungarian / Global Set Assignment (DETR-style)

Cost matrix combining class and box:

$$C_{i,j} = \alpha \, C_{\mathrm{class}}(\hat{c}_i, c_j^\star) + \beta \, C_{\mathrm{box}}(\hat{b}_i, b_j^\star).$$

Hungarian algorithm finds a minimum-cost one-to-one assignment.

Example with two predictions, two objects:

$$C = \begin{bmatrix} 0.30 & 0.80 \\ 0.40 & 0.20 \end{bmatrix}.$$

$$(1 \to 1,\ 2 \to 2) : 0.30 + 0.20 = 0.50 \quad \text{(min)}$$

$$(1 \to 2,\ 2 \to 1) : 0.80 + 0.40 = 1.20.$$

Set prediction trades NMS for a global matching objective.

Non-Maximum Suppression

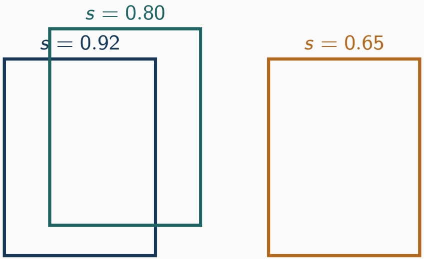
NMS: A Deterministic Rule, Not a Layer

Non-maximum suppression (per class)
Sort detections by decreasing score. Repeatedly keep the highest-scoring remaining detection d, and remove every remaining d' with $\mathrm{IoU}(b_d, b_{d'}) > \tau_{\mathrm{nms}}$.

- No neural network here — post-processing on boxes and scores.
- Pairwise overlap among kept boxes is at most τ_{nms}.
- Soft-NMS reduces scores instead of deleting.

NMS can change reported detections without changing model weights.

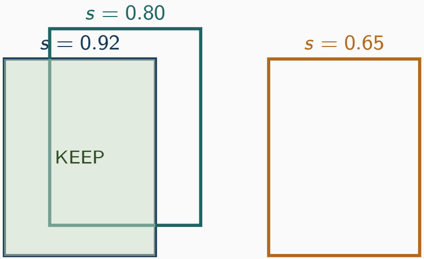
Animated: NMS on Duplicate Person Boxes



$\text{IoU}(0.92, 0.80) = 0.62, \ \text{IoU}(0.92, 0.65) = 0.10$

(1) Sort by score: $0.92, 0.80, 0.65$.

Animated: NMS on Duplicate Person Boxes

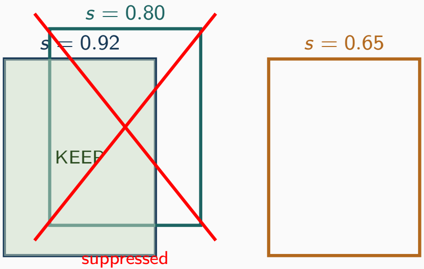


$\mathrm{IoU}(0.92, 0.80) = 0.62, \ \mathrm{IoU}(0.92, 0.65) = 0.10$

(1) Sort by score: $0.92, 0.80, 0.65$.

(2) Keep the top detection.

Animated: NMS on Duplicate Person Boxes



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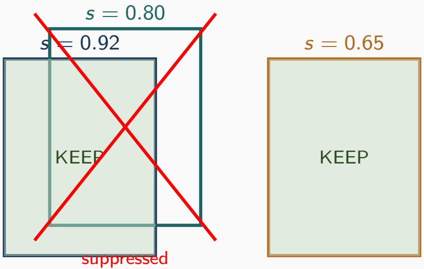
(1) Sort by score: $0.92, 0.80, 0.65$.

(2) Keep the top detection.

(3) Remove the high-IoU duplicate.

Animated: NMS on Duplicate Person Boxes



$\mathrm{IoU}(0.92, 0.80) = 0.62,\ \mathrm{IoU}(0.92, 0.65) = 0.10$

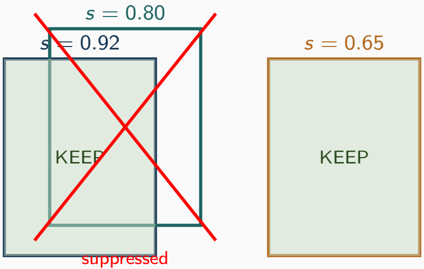
(1) Sort by score: $0.92, 0.80, 0.65$.

(2) Keep the top detection.

(3) Remove the high-IoU duplicate.

(4) Lower-IoU box survives.

Animated: NMS on Duplicate Person Boxes



$\text{IoU}(0.92, 0.80) = 0.62, \ \text{IoU}(0.92, 0.65) = 0.10$

$\tau_{\text{nms}} = 0.3$: keep 0.92, 0.65
$\tau_{\text{nms}} = 0.5$: keep 0.92, 0.65
$\tau_{\text{nms}} = 0.7$: keep all three
Aggressive vs. permissive is a validation choice.

(1) Sort by score: $0.92, 0.80, 0.65$.
(2) Keep the top detection.
(3) Remove the high-IoU duplicate.
(4) Lower-IoU box survives.
(5) Threshold sweep changes the kept set.

Low τ_{nms} helps duplicates; high helps crowded scenes.

Precision, Recall, AP, and mAP

Counts After Matching

Fix a class and IoU threshold; sort predictions by score.

- **TP**: prediction matched to a previously unmatched object.
- **FP**: unmatched, wrong class, low IoU, or duplicate.
- **FN**: ground-truth object with no matched prediction.

$$\text{precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad \text{recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}.$$

Both depend on the score threshold. Lowering it usually raises recall but can lower precision.

Detection precision/recall is computed *after* matching, not before.

Animated: Ranked Detections to PR Pairs

$(P, R) = (1.00,\ 1/3)$

Rank 1
TP

Three ground-truth objects ($m = 3$). Walking down the ranked list, precision dips at every FP, recall rises only at TPs.

Animated: Ranked Detections to PR Pairs

$(P, R) = (1.00,\ 1/3)$

$(P, R) = (1/2,\ 1/3)$

Rank 1	Rank 2
TP	FP

Three ground-truth objects ($m = 3$). Walking down the ranked list, precision dips at every FP, recall rises only at TPs.

Animated: Ranked Detections to PR Pairs

Rank 1	Rank 2	Rank 3
TP	FP	TP

$(P, R) = (1.00,\ 1/3)$

$(P, R) = (1/2,\ 1/3)$

$(P, R) = (2/3,\ 2/3)$

Three ground-truth objects ($m = 3$). Walking down the ranked list, precision dips at every FP, recall rises only at TPs.

Animated: Ranked Detections to PR Pairs

Rank 1	Rank 2	Rank 3	Rank 4
TP	FP	TP	FP

$(P, R) = (1.00,\ 1/3)$

$(P, R) = (1/2,\ 1/3)$

$(P, R) = (2/3,\ 2/3)$

$(P, R) = (1/2,\ 2/3)$

Three ground-truth objects ($m = 3$). Walking down the ranked list, precision dips at every FP, recall rises only at TPs.

Animated: Ranked Detections to PR Pairs

Rank 1 TP	Rank 2 FP	Rank 3 TP	Rank 4 FP	Rank 5 TP

$(P, R) = (1.00,\ 1/3)$

$(P, R) = (1/2,\ 1/3)$

$(P, R) = (2/3,\ 2/3)$

$(P, R) = (1/2,\ 2/3)$

$(P, R) = (3/5,\ 1)$

Three ground-truth objects ($m = 3$). Walking down the ranked list, precision dips at every FP, recall rises only at TPs.

Animated: Ranked Detections to PR Pairs

Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
TP	FP	TP	FP	TP

$(P, R) = (1.00,\ 1/3)$

$(P, R) = (1/2,\ 1/3)$

$(P, R) = (2/3,\ 2/3)$

$(P, R) = (1/2,\ 2/3)$

$(P, R) = (3/5,\ 1)$

$$\mathrm{AP}_{\mathrm{toy}} = \frac{1 + 2/3 + 3/5}{3} = \frac{34}{45} \approx 0.756$$

Three ground-truth objects ($m = 3$). Walking down the ranked list, precision dips at every FP, recall rises only at TPs.

AP rewards getting true positives early in the ranking.

Average Precision and mAP

Toy rank-average AP for one class:

$$\mathrm{AP}_{\mathrm{toy}} = \frac{1}{m}\sum_{k=1}^{K} I_k \ \mathsf{precision}_k \, .$$

Interpolated precision (for VOC-style envelopes):

$$p_{\mathrm{interp}}(r) = \max_{\tilde{r}\geq r} p(\tilde{r}).$$

Mean average precision:

$$\mathrm{mAP} = \frac{1}{|\mathsf{classes}|}\sum_{c} \mathrm{AP}_c \quad \text{(COCO: average also over IoU thresholds).}$$

- VOC: AP at $\tau = 0.5$, classic interpolation.
- COCO: average mAP over $\tau \in \{0.50, \ldots, 0.95\}$ and class.

mAP numbers compare only when split, classes, τ, and convention match.

Why a False Positive Cannot Help Precision Here

Suppose a ranked prefix has $\mathrm{TP} = t$, $\mathrm{FP} = f$, with $t + f > 0$. Add one more FP:

$$\text{precision}_{\text{new}} = \frac{t}{t + f + 1} \leq \frac{t}{t + f} = \text{precision}_{\text{old}} .$$

- Numerator stays at t.
- Denominator strictly grows.
- Recall is unchanged at that step (no new TP).

Therefore a high-confidence FP *early* in the ranking drags AP down the most.

Ranking quality matters as much as raw count of correct boxes.

Detector Families Through One Lens

Same Questions, Different Answers

Family	Candidates	Mathematical issue
Two-stage (Faster R-CNN)	Region proposals, refined boxes	Proposal quality, second-stage cls.
One-stage (YOLO, SSD, RetinaNet)	Dense boxes over locations/scales	Many negatives, loss weighting.
Anchor-free (FCOS)	Centers, point-to-side distances	Assignment without anchor shape
Transformer (DETR)	Learned object queries	Set prediction, global matching.

All four must answer: who claims the person? the bicycle? was the small sign missed? was the duplicate suppressed?

Architecture names hide shared choices about candidates, assignment, and losses.

Detector Design Checklist

At the mathematical level, every family specifies:

1. Candidate predictions (grid, proposals, queries, points)
2. Box parameterization (corner, center-size, side distances, offsets)
3. Matching rule (training assignment, evaluation matching)
4. Classification loss (CE, BCE, focal)
5. Localization loss (smooth L_1, L_1, IoU/GIoU/DIoU/CIoU)
6. Post-processing (NMS variant, set-based, or none)
7. Evaluation metric (VOC AP, COCO mAP, etc.)

A gain in mAP comes from one of these slots — name which one.

Failure Modes and Diagnostics

Error Categories: Diagnose Before You Redesign

Error type	Symptom	First check
Localization	Right class, low IoU	Compare AP at 0.50 vs. 0.75
Classification	Tight box, wrong class	Class confusion examples
Duplicate	Several boxes per object	Sweep τ_{nms} on val
Missed small object	No matched prediction	Break AP by object size
Background FP	High score on no object	Hard negatives, calibration
Score miscalibration	Top scores not most reliable	Calibration / temperature

Detector-level mAP is a summary, not a diagnosis.

Debug detections by error type before changing the whole model.

Diagnostic Example: AP at Two IoU Thresholds

Suppose a bicycle detector reports:

$$\mathrm{AP}_{\tau=0.50} = 0.78, \qquad \mathrm{AP}_{\tau=0.75} = 0.41.$$

Reading:

- The model often finds the right object (high AP at 0.5).
- Boxes are consistently loose (large drop at 0.75).

First responses:

- Inspect box targets and image-resize math.
- Check anchor shapes vs. actual object widths/heights.
- Revisit localization loss (smooth L_1 vs. IoU-family).
- *Do not* immediately add classes or a bigger classification head.

Geometry symptoms call for geometry fixes, not more classification capacity.

Applied Checklist Before Reporting

Before reporting a detector:

- Fix the metric convention to match the target benchmark.
- Inspect FPs and FNs by class.
- Compare AP across IoU thresholds, not just at 0.5.
- Break results down by object size when possible.
- Tune score and NMS thresholds on validation, not on test.
- Verify coordinate transforms after every resize / crop / pad.
- Only compare numbers computed on the same split.

Model selection should be tied to a ranked set of class-labeled geometric predictions.

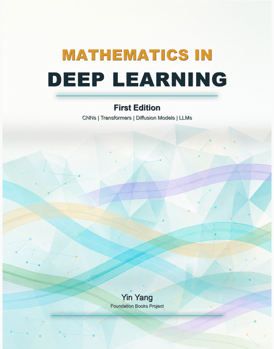
Summary

Chapter in One Slide

- **Boxes:** corner and center-size forms; coordinate convention is part of the data format.
- **IoU:** geometric overlap score in $[0, 1]$, with a benchmark-specific threshold τ.
- **Detection = cls + loc:** multitask loss $\mathcal{L}_{\text{cls}} + \lambda\mathcal{L}_{\text{box}}$.
- **Anchors / dense prediction:** candidates everywhere, most of them negative; assignment thresholds shape the training set.
- **Matching:** many positives at training, one-to-one at evaluation; duplicates become FPs.
- **NMS:** deterministic post-processing; tuned, not learned.
- **Precision / Recall / AP / mAP:** ranking metrics computed *after* matching.
- **Families:** two-stage, one-stage, anchor-free, transformer — same checklist of design slots.
- **Diagnostics:** separate error types before architectural changes.

Next: from detection to dense prediction — segmentation, masks, and per-pixel evaluation.

Free Chapter Overview Slides



Mathematics in Deep Learning

Chapter overview slides for the book by Yin Yang.

Book page	foundation-books.github.io/mathematics-in-deep-learning
Code	github.com/foundation-books/mathematics-in-deep-learning-code
Paperback	amazon.com/dp/B0H1ZZHDT2
Hardcover	amazon.com/dp/B0H1ZL11MJ
Kindle	amazon.com/dp/B0GX32M8SP
License	CC BY-NC-ND 4.0

 Full explanations, exercises, and companion-code context are in the book and linked repository.